

Job Offer Comparison Tool - Recreate Environment Steps

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1. Dependencies

- Node.js: 26.5.1
- MySQL/MariaDB: 8/11.7.2
- Redis: 8.8.0
- MongoDB: 8.3.7

2. Code

```
git clone https://code.vt.edu/nchatpol/chatpolarak_nathan_final_webapp.git
cd chatpolarak_nathan_final_webapp
```

Run all commands from that directory.

3. Create the MySQL database and its user

Log in to MySQL as an administrator and run:

```
CREATE USER 'joboffers'@'localhost' IDENTIFIED BY 'hokies2026';
CREATE DATABASE job_offers CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
GRANT ALL PRIVILEGES ON job_offers.* TO 'joboffers'@'localhost';
FLUSH PRIVILEGES;
```

Step 5 already expects this password.

4. Load the data

Two ways to do this. Pick one.

Option A starts empty: tables plus the reference data (states, the 12 cities, federal and state tax brackets). No accounts, no offers.

```
mysql -h 127.0.0.1 -u joboffers -p job_offers < db/mysql/schema.sql
mysql -h 127.0.0.1 -u joboffers -p job_offers < db/mysql/seed.sql
mongosh job_offers db/mongo/init.js
```

Option B restores the submitted dumps instead. Same tables, but with the sample account and its offers and comparisons already in there.

```
mysql -h 127.0.0.1 -u joboffers -p job_offers < db/dumps/job_offers_mysql.sql
mongorestore db/dumps/mongo
```

5. Configure the server

```
cp server/.env.example server/.env
```

Nothing in `server/.env` needs to change if the databases are all local and on their default ports. The copied defaults already match step 3.

Setting	Default	What it is
PORT	3000	Port the application is served on
MYSQL_HOST	127.0.0.1	MySQL host
MYSQL_PORT	3306	MySQL port
MYSQL_USER	joboffers	The user created in step 3
MYSQL_PASSWORD	hokies2026	Matches step 3, already set in <code>server/.env.example</code>
MYSQL_DATABASE	job_offers	Database name
REDIS_URL	redis://127.0.0.1:6379	Redis connection
MONGO_URL	mongodb://127.0.0.1:27017	MongoDB connection
MONGO_DATABASE	job_offers	MongoDB database name
CLIENT_ORIGIN	http://localhost:5173	Only used by the development setup in section 9

`server/.env` holds a password, so it is not in the repository.

6. Install the dependencies

```
cd server && npm install
cd ../client && npm install
```

7. Build the front end and start the server

```
cd client && npm run build
cd ../server && npm start
```

`client/dist` isn't in the repository, so this step can't be skipped. The server serves that folder. The whole app ends up at:

```
http://localhost:3000
```

On startup the server prints a line for each database it connects to and a line saying where the client build was found. If any database is unreachable it says which one and stops.